



Firewall Administration

FORTINET FORTIGATE NGFW DEPLOYMENT & SYSTEM CONFIGURATION REPORT



Appliance Model : FortiGate-VM64 (FortiOS v7.6.7)

Deployment Platform : VMware Workstation Hypervisor Infrastructure

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1. Executive Summary & Environment Specifications

This document outlines the initial deployment, system initialization, and CLI environment verification of the FortiGate Next-Generation Firewall (NGFW) within a virtualized lab infrastructure.

Appliance Type: FortiGate Virtual Machine (FortiGate-VM64)

Firmware Version: FortiOS v7.6.7 (Build 3704)

System Architecture: x86_64 Virtual Appliance

Hypervisor Platform: VMware Workstation Pro

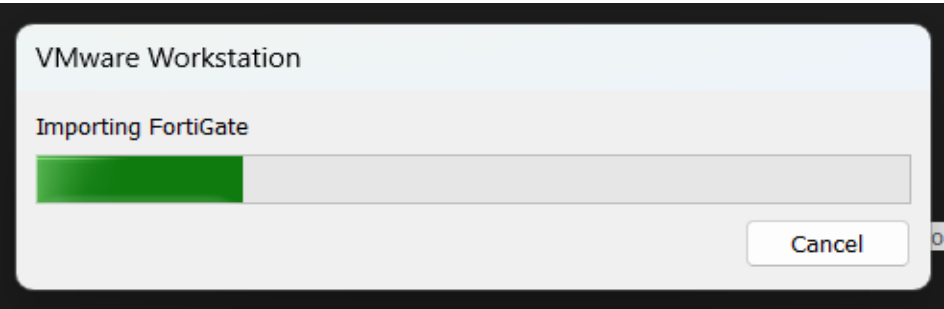
Serial Number: FGVMEVR3EICHXJ62

2. Virtual Machine Deployment & Disk Initialization

Step 2.1: Importing Virtual Machine Template

The deployment process was initiated by importing the Open Virtualization Format (FortiGate-VM64.ovf) template into VMware Workstation. This predefines the virtual hardware allocation (vCPU, RAM, and Network Adapters) required for FortiOS.

FortiGate-VM64.nsxt.ovf	9/12/2026 6:55 PM	Open Virtualizatio...	14 KB
FortiGate-VM64.ovf	9/12/2026 6:55 PM	Open Virtualizatio...	27 KB
FortiGate-VM64.scoreboard	9/13/2026 11:56 PM	SCOREBOARD File	8 KB



Step 2.2: First Boot & Storage Partitioning

Upon initial boot, FortiOS loaded the kernel files (flatkc and rootfs.gz). The system automatically formatted and partitioned the virtual disks (/dev/sda and /dev/sdb) to prepare log partitions (LOGUSEDXCF9B9583) before executing an automated system reboot.

```
Loading flatkc... ok
Loading /rootfs.gz...ok

System is starting...
Formatting shared data partition ... done!
Starting system maintenance...
Scanning /dev/sda1... (100%)
Scanning /dev/sda2... (100%)
Serial number is FGUMEUR3EICHXJ62

Disk usage changed, please wait for reboot...

Formatting the disk...
Unmounting...
Partitioning and formatting /dev/sdb label LOGUSEDXCF9B9583 ... done

The system is going down NOW !!
```

3. Administrator Access & Password Policy Enforcement

Step 3.1: Default Authentication

Initial system access was established via the VMware Console using the default administrative credentials (admin with no password).

```
Loading flatkc... ok
Loading /rootfs.gz...ok

System is starting...
Serial number is FGUMEUR3EICHXJ62

FortiGate-UM64 login: admin
Password:
Verifying password...

You are forced to change your password. Please input a new password.
According to the password policy enforced on this device, please change your password!
New password must conform to the following policy:
minimum-length=12 upper-case-letter=1 lower-case-letter=1 number=1 non-alphanumeric=1

New Password: _
```

Step 3.2: Mandatory Password Policy Enforcement

FortiOS v7.6 enforced a mandatory password update upon first login to align with enterprise security standards. The new password was required to meet the following parameters:

- **Minimum Length:** 12 characters
- **Complexity Requirements:** At least 1 uppercase letter, 1 lowercase letter, 1 numerical digit, and 1 non-alphanumeric character.

Step 3.3: Successful CLI Session Verification

Following authentication, full administrative CLI privileges were granted, enabling network interface configuration and service provisioning.

```
FGUMEVQAOW0KOI83 login: admin
Password:
Verifying password...

Welcome!

FGUMEVQAOW0KOI83 #
FGUMEVQAOW0KOI83 #
```

4. CLI System Inspection & Command Breakdown

Step 4.1: Interface Status Verification (show system interface ?)

Executing the show system interface command retrieved the initial network configuration:

- Port1 (WAN/Management): Configured via DHCP, receiving the dynamic IP address 192.168.233.135/24.

```
FGUMEVQAOW0KOI83 login: admin
Password:
Verifying password...

Welcome!

FGUMEVQAOW0KOI83 # sh system interface
name      Name.
default-mesh static  0.0.0.0 0.0.0.0 0.0.0.0 0.0.0.0 up  disable vap-swi
tch
fortilink static  0.0.0.0 0.0.0.0 10.255.1.1 255.255.255.0 up  disable a
ggregate
l2t.root static  0.0.0.0 0.0.0.0 0.0.0.0 0.0.0.0 up  disable tunnel
naf.root static  0.0.0.0 0.0.0.0 0.0.0.0 0.0.0.0 up  disable tunnel
port1 dhcp  0.0.0.0 0.0.0.0 192.168.233.135 255.255.255.0 up  disable ph
ysical
port2 static  0.0.0.0 0.0.0.0 0.0.0.0 0.0.0.0 up  disable physical
port3 static  0.0.0.0 0.0.0.0 0.0.0.0 0.0.0.0 up  disable physical
ssl.root static  0.0.0.0 0.0.0.0 0.0.0.0 0.0.0.0 up  disable tunnel
```

Step 4.2: System Performance Monitoring (get system performance status)

Real-time resource utilization was verified to ensure system stability prior to traffic routing:

- **CPU Idle State:** 100% idle (0% user / system utilization)
- **RAM Consumption:** 47.5% used of 2.0 GB total allocation
- **Active Sessions:** 3 active sessions recorded

```
FGVMEVQAOW0KOI83 # get system performance status_

CPU states: 0% user 0% system 0% nice 100% idle 0% iowait 0% irq 0% softirq
CPU0 states: 0% user 0% system 0% nice 100% idle 0% iowait 0% irq 0% softirq
Memory: 2041876k total, 970612k used (47.5%), 696672k free (34.1%), 374592k free
able (18.4%)
Average network usage: 0 / 0 kbps in 1 minute, 0 / 0 kbps in 10 minutes, 0 / 3 k
bps in 30 minutes
Maximal network usage: 5 / 1 kbps in 1 minute, 5 / 2 kbps in 10 minutes, 105 / 8
84 kbps in 30 minutes
Average sessions: 3 sessions in 1 minute, 5 sessions in 10 minutes, 5 sessions i
n 30 minutes
Maximal sessions: 4 sessions in 1 minute, 11 sessions in 10 minutes, 13 sessions
in 30 minutes
Average session setup rate: 0 sessions per second in last 1 minute, 0 sessions p
er second in last 10 minutes, 0 sessions per second in last 30 minutes
Maximal session setup rate: 0 sessions per second in last 1 minute, 2 sessions p
er second in last 10 minutes, 4 sessions per second in last 30 minutes
Average NPU sessions: 0 sessions in last 1 minute, 0 sessions in last 10 minutes
, 0 sessions in last 30 minutes
Maximal NPU sessions: 0 sessions in last 1 minute, 0 sessions in last 10 minutes
, 0 sessions in last 30 minutes
Virus caught: 0 total in 1 minute
IPS attacks blocked: 0 total in 1 minute
Uptime: 0 days, 6 hours, 21 minutes

FGVMEVQAOW0KOI83 #
```

Step 4.3: Comparison of Core FortiOS CLI Commands

Command	Data Type Displayed	Purpose & Scope
show	Configuration File	Displays static configurations defined by the administrator in the configuration file.
get	Operational / Runtime	Displays active operational status, including dynamically assigned values (e.g., DHCP IPs).
diagnose	Troubleshooting	Used for real-time packet tracing, kernel debugging, and deep system diagnostics.

Step 4.4: Low-Level IP Address Diagnostics (diagnose ip address list)

The `diagnose ip address list` command was executed to perform a low-level kernel query to list all active IPv4 addresses bound to physical and logical interfaces.

```
FGUMEVQAOW0KOI83 # diagnose ip address list
IP=192.168.233.135->192.168.233.135/255.255.255.0 index=3 devname=port1
IP=127.0.0.1->127.0.0.1/255.0.0.0 index=14 devname=root
IP=10.255.1.1->10.255.1.1/255.255.255.0 index=17 devname=fortilink
IP=127.0.0.1->127.0.0.1/255.0.0.0 index=19 devname=vsys_ha
IP=127.0.0.1->127.0.0.1/255.0.0.0 index=21 devname=vsys_fgfm
FGUMEVQAOW0KOI83 #
```

Key Technical Observations:

- **port1 (Primary Management Interface):** Bound to 192.168.233.135/24 (Index 3).
- **fortilink (FortiSwitch Integration Interface):** Pre-configured with static IP 10.255.1.1/24 (Index 17).
- **Logical & Internal Interfaces (root, vsys_ha, vsys_fgfm):** Assigned local loopback addresses (127.0.0.1) for internal inter-process communications, High Availability (HA) signaling, and FortiManager (FGFM) daemon operations.

Step 4.5: Routing Table Verification (get router info routing-table all)

The `get router info routing-table all` command was executed to inspect the active Routing Information Base (RIB) within the default virtual routing domain (VRF=0).

```
FGUMEVQAOW0KOI83 # get router info routing-table all
Codes: K - kernel, C - connected, S - static, R - RIP, B - BGP
       O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       U - BGP UPv4
       * - candidate default

Routing table for VRF=0
S*   0.0.0.0/0 [5/0] via 192.168.233.2, port1, [1/0]
C     192.168.233.0/24 is directly connected, port1

FGUMEVQAOW0KOI83 #
```

Key Routing Entries Breakdown:

- **Default Route (S* 0.0.0.0/0):** Automatically provisioned via DHCP on port1. Traffic destined for external networks is forwarded to Next-Hop Gateway 192.168.233.2 with an Administrative Distance (AD) of 5 and Metric of 0.
- **Connected Subnet (C 192.168.233.0/24):** Represents the directly connected local broadcast domain assigned to physical interface port1.

Step 4.6: Kernel Forwarding Information Base Inspection (get router info kernel)

The get router info kernel command was executed to inspect the active Forwarding Information Base (FIB) programmed directly into the FortiOS Linux kernel. While the RIB manages routing protocols, the FIB represents the actual data-plane table used for hardware/software packet forwarding decisions.

```
FGUMEVQAOW0KOI83 # get router info kernel
```

```
tab=65535 vf=0 vrf=0 scope=253 type=3 proto=2 prio=0 0.0.0.0/0.0.0.0/0->10.255.1
.0/32 pref=10.255.1.1 gwy=0.0.0.0 dev=17(fortilink)
tab=65535 vf=0 vrf=0 scope=254 type=2 proto=2 prio=0 0.0.0.0/0.0.0.0/0->10.255.1
.1/32 pref=10.255.1.1 gwy=0.0.0.0 dev=17(fortilink)
tab=65535 vf=0 vrf=0 scope=253 type=3 proto=2 prio=0 0.0.0.0/0.0.0.0/0->10.255.1
.255/32 pref=10.255.1.1 gwy=0.0.0.0 dev=17(fortilink)
tab=65535 vf=0 vrf=0 scope=253 type=3 proto=2 prio=0 0.0.0.0/0.0.0.0/0->127.0.0.
0/32 pref=127.0.0.1 gwy=0.0.0.0 dev=14(root)
tab=65535 vf=0 vrf=0 scope=254 type=2 proto=2 prio=0 0.0.0.0/0.0.0.0/0->127.0.0.
0/8 pref=127.0.0.1 gwy=0.0.0.0 dev=14(root)
tab=65535 vf=0 vrf=0 scope=254 type=2 proto=2 prio=0 0.0.0.0/0.0.0.0/0->127.0.0.
1/32 pref=127.0.0.1 gwy=0.0.0.0 dev=14(root)
tab=65535 vf=0 vrf=0 scope=253 type=3 proto=2 prio=0 0.0.0.0/0.0.0.0/0->127.255.
255/32 pref=127.0.0.1 gwy=0.0.0.0 dev=14(root)
tab=65535 vf=0 vrf=0 scope=253 type=3 proto=2 prio=0 0.0.0.0/0.0.0.0/0->192.168.
233.0/32 pref=192.168.233.135 gwy=0.0.0.0 dev=3(port1)
tab=65535 vf=0 vrf=0 scope=254 type=2 proto=2 prio=0 0.0.0.0/0.0.0.0/0->192.168.
233.135/32 pref=192.168.233.135 gwy=0.0.0.0 dev=3(port1)
tab=65535 vf=0 vrf=0 scope=253 type=3 proto=2 prio=0 0.0.0.0/0.0.0.0/0->192.168.
233.255/32 pref=192.168.233.135 gwy=0.0.0.0 dev=3(port1)
tab=65534 vf=0 vrf=0 scope=0 type=1 proto=11 prio=1 0.0.0.0/0.0.0.0/0->0.0.0.0/0
pref=0.0.0.0 gwy=192.168.233.2 dev=3(port1)
tab=65534 vf=0 vrf=0 scope=253 type=1 proto=2 prio=0 0.0.0.0/0.0.0.0/0->10.255.1
.0/24 pref=10.255.1.1 gwy=0.0.0.0 dev=17(fortilink)
--More--
```

Key Kernel Table Breakdown:

- **Active Forwarding Route (tab=65534):** Contains the active default gateway entry (0.0.0.0/0) directing traffic out of interface dev=3(port1) toward next-hop 192.168.233.2.
- **Local Interface Binding (tab=65535):** Contains host-route entries (/32) for interface IP addresses (e.g., 192.168.233.135/32 and 10.255.1.1/32) required for administrative control-plane handling and local packet termination.

Step 4.7: Internet Connectivity & DNS Resolution Testing (execute ping)

The execute ping google.com command was executed from the CLI to verify outbound WAN reachability and Domain Name System (DNS) resolution capability directly from the FortiGate control plane.

```
FGUMEVQAOW0KOI83 # execute ping google.com
PING google.com (142.251.39.174): 56 data bytes
64 bytes from 142.251.39.174: icmp_seq=0 ttl=128 time=46.2 ms
64 bytes from 142.251.39.174: icmp_seq=1 ttl=128 time=44.7 ms
64 bytes from 142.251.39.174: icmp_seq=2 ttl=128 time=48.1 ms
64 bytes from 142.251.39.174: icmp_seq=3 ttl=128 time=45.2 ms
64 bytes from 142.251.39.174: icmp_seq=4 ttl=128 time=45.3 ms

--- google.com ping statistics ---
5 packets transmitted, 5 packets received, 0% packet loss
round-trip min/avg/max = 44.7/45.9/48.1 ms

FGUMEVQAOW0KOI83 #
```

Key Diagnostic Results:

- **DNS Resolution Success:** The system successfully resolved the domain google.com to public IPv4 address 142.251.39.174.
- **Outbound Reachability:** Recorded 0% packet loss across 5 ICMP Echo requests with an average Round-Trip Time (RTT) of 45.9 ms.
- **Verification Outcome:** Confirms active Internet connectivity through port1 via the assigned default gateway (192.168.233.2).

Step 4.8: DNS Proxy Cache & FQDN Inspection (diagnose test application dnsproxy 6)

The diagnose test application dnsproxy 6 command was executed to evaluate the active FQDN (Fully Qualified Domain Name) cache entries stored within the FortiOS DNS Proxy daemon (dnsproxy).

```
FGUMEVQAOW0KOI83 #
FGUMEVQAOW0KOI83 # diagnose test application dnsproxy 6_
```

```

ttl=300:133:133) 40.126.53.13 (ttl=300:133:133) 40.126.53.15 (ttl=300:133:133)
20.231.128.66 (ttl=300:133:133) 40.126.53.14 (ttl=300:133:133) 20.231.128.6
5 (ttl=300:133:133) 20.190.177.148 (ttl=300:188:188) 20.190.177.20 (ttl=300:188:
188) 20.190.177.147 (ttl=300:188:188)
20.190.177.23 (ttl=300:188:188) 20.190.177.149 (ttl=300:188:188) 20.190.177
.22 (ttl=300:188:188) 20.190.147.0 (ttl=300:188:188) 20.190.177.84 (ttl=300:188:
188) 40.126.31.3 (ttl=288:231:231)
20.190.159.129 (ttl=288:231:231) 20.190.159.23 (ttl=288:231:231) 20.190.159
.2 (ttl=288:231:231) 40.126.31.129 (ttl=288:231:231) 20.190.159.131 (ttl=288:231
:231) 40.126.31.73 (ttl=288:231:231)
40.126.31.1 (ttl=288:231:231)
name=*.dropbox.com ver=IPv4 wait_list=0 timer=0 min_ttl=0 cache_ttl=0 slot=-1 nu
m=0 wildcard=1
name=login.windows.net ver=IPv4 wait_list=0 timer=51 min_ttl=123 cache_ttl=0 slo
t=-1 num=13 wildcard=0
20.190.181.0 (ttl=300:57:57) 40.126.53.12 (ttl=300:57:57) 40.126.53.15 (ttl
=300:57:57) 40.126.53.13 (ttl=300:57:57) 40.126.53.7 (ttl=300:57:57)
20.190.181.4 (ttl=300:234:234) 20.190.181.6 (ttl=300:234:234) 40.126.53.10
(ttl=300:234:234) 40.126.53.11 (ttl=300:234:234) 40.126.53.14 (ttl=300:234:234)
40.126.53.16 (ttl=300:234:234)
40.126.53.9 (ttl=300:234:234) 20.190.181.5 (ttl=300:234:234)

FQDN num=6

FGUMEVQAOW0KOI83 # _

```

Key Diagnostic Results:

- **FQDN Object Resolution:** Demonstrates active IP resolution for system domain objects such as login.windows.net and wildcard objects like *.dropbox.com.
- **DNS Caching Mechanism:** Displays cached IPv4 address lists alongside Time-To-Live (ttl) values, ensuring optimized packet processing and rapid policy enforcement for FQDN-based firewall rules.
- **Cached Objects Count:** Confirmed FQDN num=6 active domain entries maintained in the proxy memory structure.

Step 4.9: Address Resolution Protocol (ARP) Cache Verification (get system arp)

The get system arp command was executed to inspect the active Layer 2 to Layer 3 address mappings maintained within the FortiOS system ARP cache.

```

FGUMEVQAOW0KOI83 # get system arp
Address      Age(Min)    Hardware Addr    Interface
192.168.233.1 31          00:50:56:c0:00:08 port1
192.168.233.2 0           00:50:56:fc:df:df port1

```

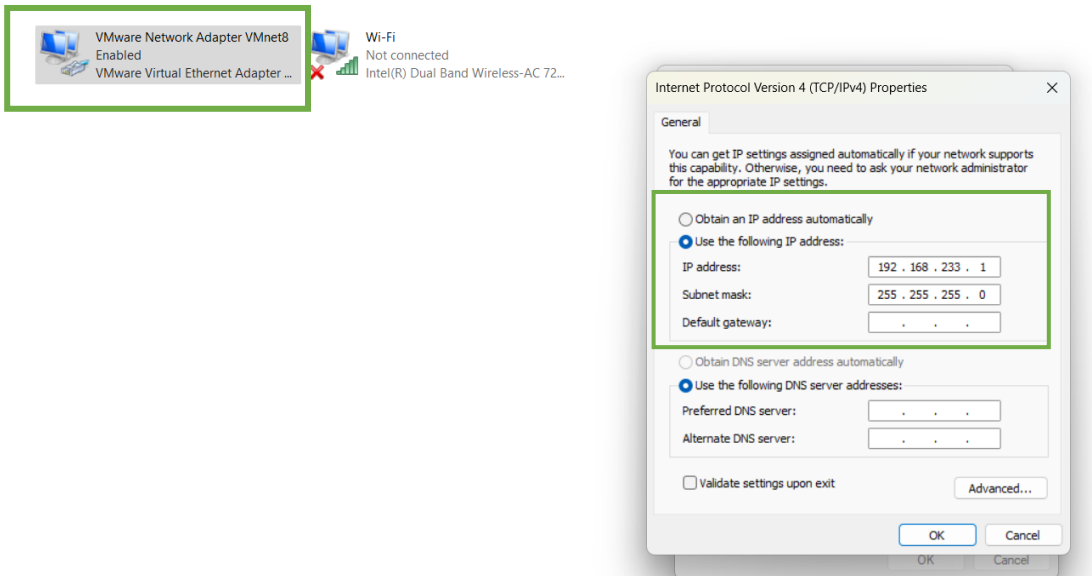
Key ARP Entries Analysis:

- **VMware Virtual Host Interface (192.168.233.1):** Mapped to MAC address 00:50:56:c0:00:08 via port1 with an entry age of 31 minutes.
- **VMware Virtual Gateway (192.168.233.2):** Mapped to MAC address 00:50:56:fc:df:df via port1 with an entry age of 0 minutes, reflecting recent control-plane ping interactions.
- **Verification Outcome:** Confirms successful Layer 2 neighbor discovery and operational ARP table construction across the port1 broadcast domain.

5. Web GUI Provisioning & Host Connectivity Integration

Step 5.1: Network Subnet Alignment & Host Adapter Verification

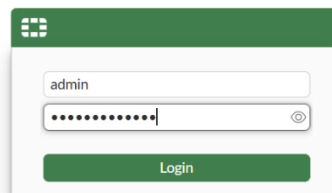
To establish HTTPS Graphical User Interface (GUI) management access, the host machine's virtual network interface (VMware Network Adapter VMnet8) was aligned with the FortiGate management subnet.



- **Host Machine IP (VMnet8):** 192.168.233.1 (Subnet Mask: 255.255.255.0)
- **FortiGate Management Interface (port1):** 192.168.233.135/24 (Dynamically assigned via VMware DHCP)
- **Layer 2 / Layer 3 Reachability:** Confirmed via ARP table resolution between host MAC 00:50:56:c0:00:08 and FortiGate port1.

Step 5.2: HTTPS Administrative Authentication

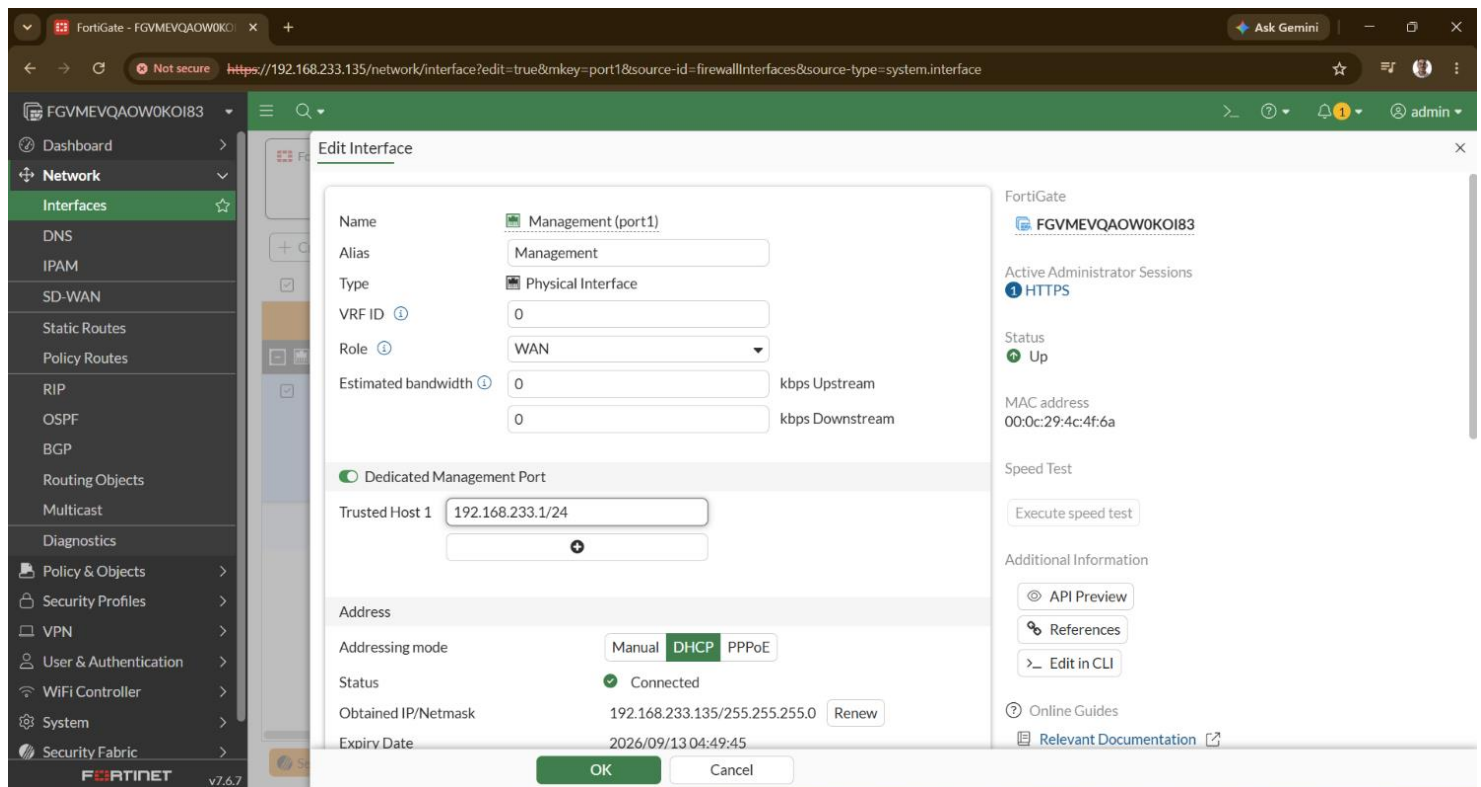
With Layer 3 connectivity established across the 192.168.233.0/24 broadcast domain, HTTPS traffic was initiated toward (<https://192.168.233.135>).



- **Authentication Endpoint:** FortiGate Secure Login Web Portal.
- **Administrative Credentials:** Authenticated using admin alongside the updated mandatory compliance password.
- **Control Plane Access:** Successfully redirected to the FortiOS Graphical User Interface (GUI) dashboard.

Step 5.3: Interface Customization & Security Rating Remediation

1. **Initial Interface Customization:** port1 was edited to assign the custom alias Management (displaying as Management (port1)) for control-plane identification.



2. **Security Insights Identification:** Upon reviewing the updated interface state in **Network > Interfaces**, the FortiOS Security Rating engine generated two active policy warnings:

- **Unsecure Protocol - HTTP:** Flagged unencrypted HTTP administrative access as a credential exposure risk.
- **"WAN" Classified Interfaces with Management Access:** Flagged active administrative services (HTTP, SNMP) on an interface assigned to the WAN role.

Administrative services (SNMP)

The screenshot shows the FortiGate web interface for configuration. The left sidebar contains the navigation menu with 'Network' > 'Interfaces' selected. The main content area is titled 'FortiGate VM64' and shows a calendar for the month of January. Below the calendar, there is a '+ Create new' button and a search bar. The 'Administrative Access' tab is selected, showing a table of administrative access configurations. The table has columns for Name, Type, Members, IP/Netmask, and Administrative access. The first row is for 'Management (port1)' under the 'Physical Interface' type, with IP/Netmask '192.168.233.135/255.255.255.0'. The 'Administrative access' column shows a list of protocols: PING, HTTPS, SSH, and SNMP. The 'SNMP' protocol is highlighted with a green box. Below the table, there is a 'Security Rating Insights' section with a green box around the 'Security Rating Insights (2)' link. The bottom status bar shows 'FORTINET v7.6.7'.

Name	Type	Members	IP/Netmask	Administrative access
Management (port1)	Physical Interface		192.168.233.135/255.255.255.0	PING HTTPS SSH SNMP

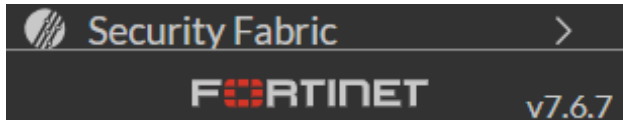
Unsecure Protocol (HTTP)

The screenshot shows the FortiGate web interface for configuration. The left sidebar contains the navigation menu with 'Network' > 'Interfaces' selected. The main content area is titled 'FortiGate VM64' and shows a calendar for the month of January. Below the calendar, there is a '+ Create new' button and a search bar. The 'Unsecure Protocol - HTTP' tab is selected, showing a table of unsecure protocol configurations. The table has columns for Name, Type, Members, IP/Netmask, and Administrative access. The first row is for 'Management (port1)' under the 'Physical Interface' type, with IP/Netmask '192.168.233.135/255.255.255.0'. The 'Administrative access' column shows a list of protocols: PING, HTTPS, SSH, and SNMP. The 'SNMP' protocol is highlighted with a green box. Below the table, there is a 'Security Rating Insights' section with a green box around the 'Security Rating Insights (2)' link. The bottom status bar shows 'FORTINET v7.6.7'.

Name	Type	Members	IP/Netmask	Administrative access
Management (port1)	Physical Interface		192.168.233.135/255.255.255.0	PING HTTPS SSH SNMP

3. **Hardening & Risk Mitigation:** A comprehensive edit was applied to port1 to remediate both identified vulnerabilities:

- **Insecure Service Removal:** Disabled clear-text HTTP and unnecessary SNMP services, restricting protocols strictly to encrypted HTTPS, SSH, and control PING.
- **Access Hardening (Trusted Host):** Enabled the Dedicated Management Port profile and bound management access exclusively to the authorized management subnet (192.168.233.1/24).



Step 5.4: Post-Remediation Verification & Final Control-Plane State

Following the administrative hardening and security remediation process, the **Network > Interfaces** panel was inspected to verify the final operational state across interfaces:

	+ Create new	🔍 Search							
	Name	Type	Members	IP/Netmask	Administrative access				
802.3ad Aggregate 1									
☐	fortilink	802.3ad Aggregate		Dedicated to FortiSwitch	PING Security Fabric Connection				
Physical Interface 3									
☐	Management (port1)	Physical Interface		192.168.233.135/255.255.255.0	PING HTTPS				
☒	Wan (port2)	Physical Interface		192.168.1.37/255.255.255.0	PING				

- **Security Warning Resolution:** Confirmed that all active FortiOS Security Rating warnings including the unsecure HTTP protocol alert and the WAN-classified management access risk were fully resolved and cleared from the dashboard.
- **Management Interface Baseline:** Verified that Management (port1) is operating securely with only authorized administrative protocols (PING, HTTPS) enabled over 192.168.233.135/24.
- **Data-Plane WAN Provisioning:** Confirmed the initialization and active binding of Wan (port2) with IP subnet 192.168.1.37/24, configured strictly with PING response for connectivity verification.
- **Access Control Verification:** Confirmed that out-of-band management access remains strictly constrained by the Dedicated Management Port policy bound to the trusted host subnet.
- **Hardware Topology Representation:** Confirmed clean visual and operational mapping across physical interfaces (port1 and port2 active).

6: Administrative Accounts & Access Control Profiles (RBAC)

To enforce the principle of least privilege and establish granular role-based access control (RBAC), custom administrative profiles and dedicated user accounts were configured on the FortiGate unit.

Step 6.1: Custom Admin Profile Provisioning (Customized profile)

Rather than granting full operational control (super_admin) to all operators, a restricted Access Control Profile named Customized profile was defined under System > Admin Profiles to enforce granular privilege management:

- **Firewall Policy & Objects Access:** Configured with **Read** permissions for security policies while restricting modification capabilities.
- **User & Device Management:** Assigned **Read** permissions to view authenticated entity profiles.
- **Administrator Users Management:** Set strictly to **Read** to prevent unauthorized privilege escalation or modification of other admin accounts.
- **System & Network Configurations:** Assigned selective **Read/Write** control over routine operational parameters.

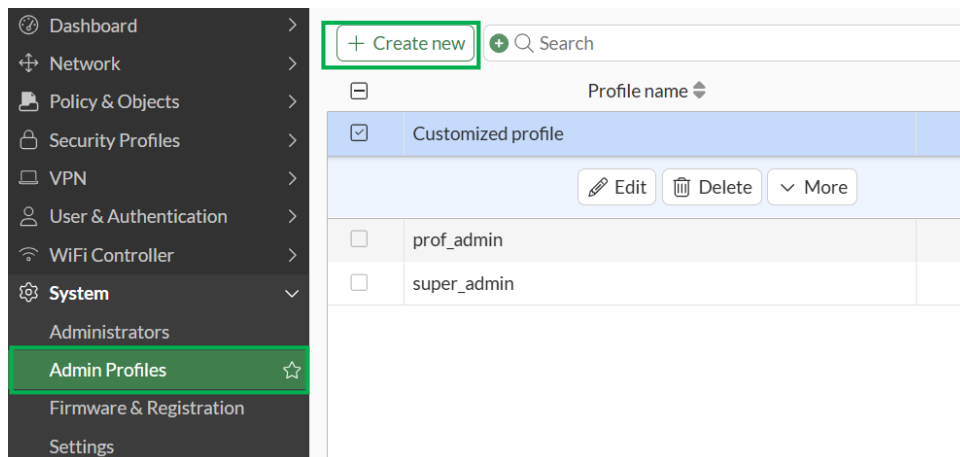


Figure 6.1: Creation and Provisioning of the Custom Access Control Profile (Customized profile) under System > Admin Profiles.

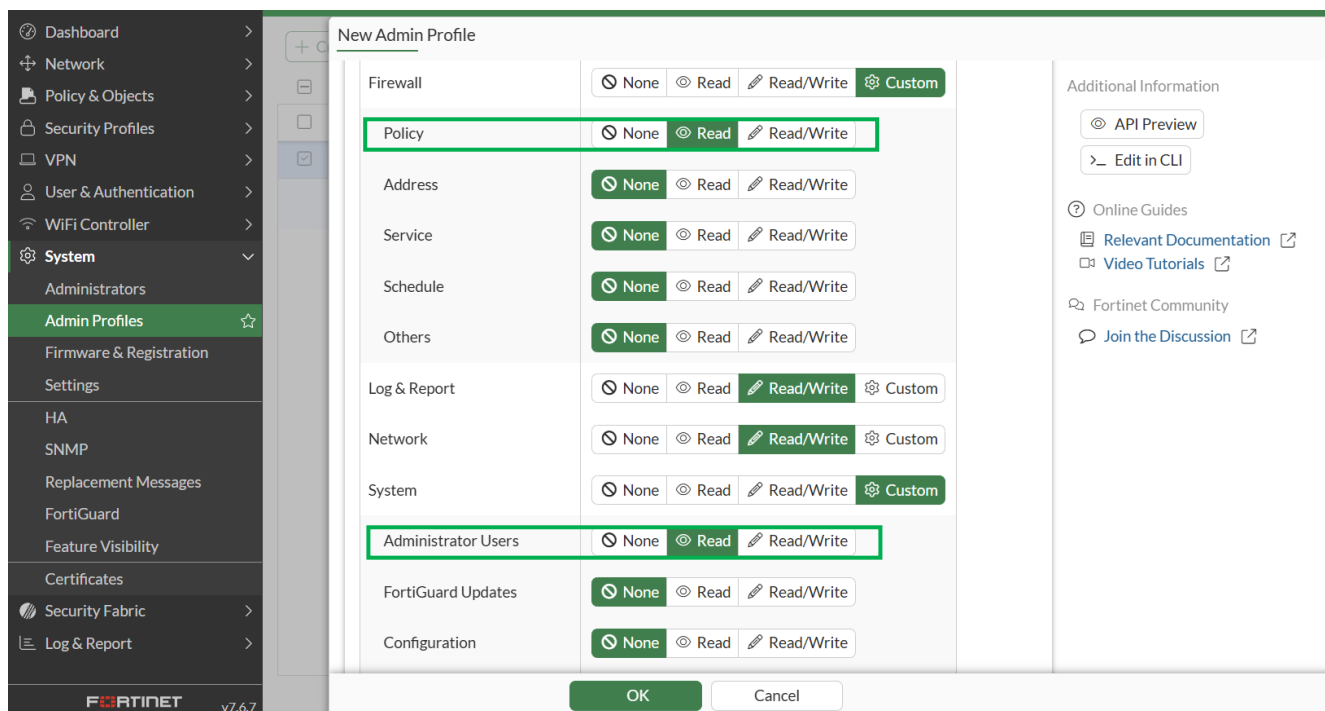
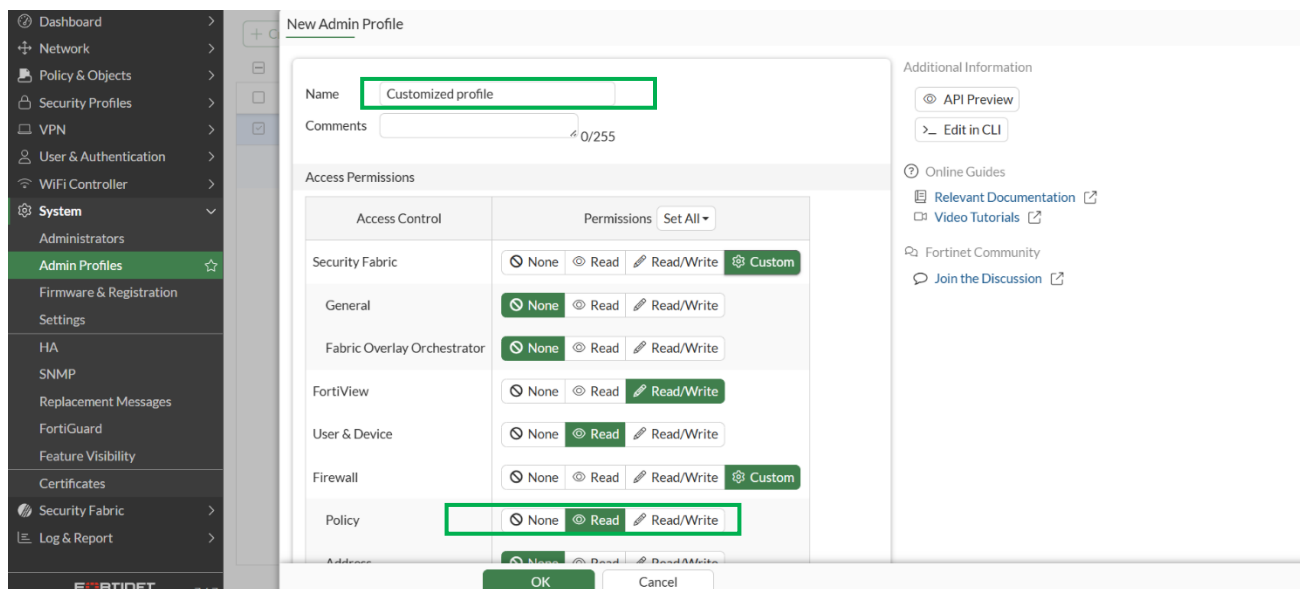


Figure 6.2: Configuring Read Only and Custom Access Permissions for Policy and Administrator Users under Customized profile.

Step 6.2: Administrator Account Creation & Tiered Privilege Mapping

Under **System > Administrators**, three local administrative accounts were registered to isolate system duties:

1. **admin (System Account)**: Default built-in system administrator account assigned to the **super_admin** profile with unrestricted administrative capabilities.
2. **Mahmoud_Hamza (Full Administrator Account)**: Provisioned as a primary local administrative account bound to the **super_admin** privilege level for system-wide configuration and maintenance.
3. **Mennahtullah (Restricted Operations Administrator)**: Provisioned as a restricted operator account bound strictly to the **Customized profile**.

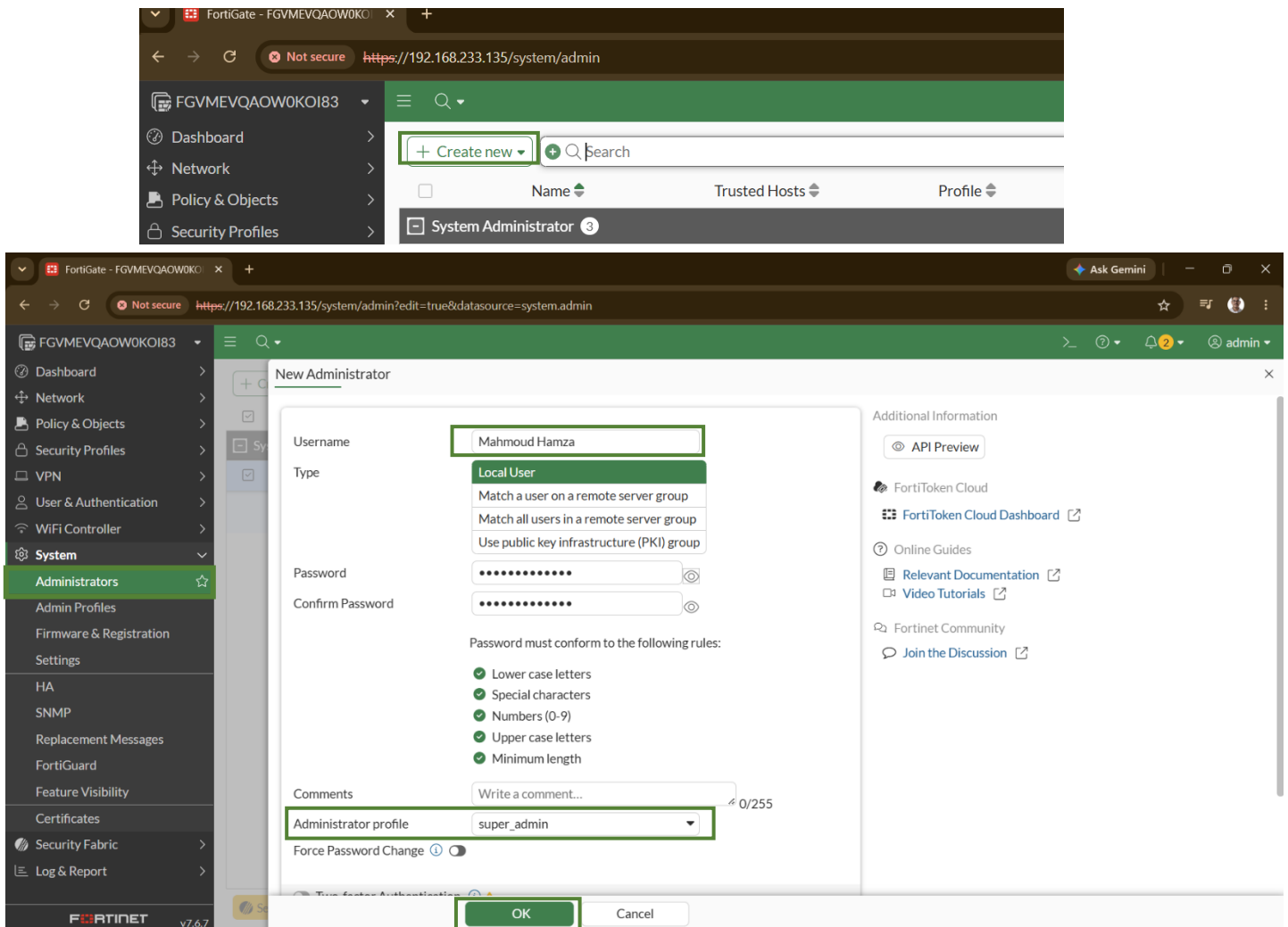


Figure 6.3: Initiating New Administrator Creation and Assigning super_admin Profile to Mahmoud_Hamza.

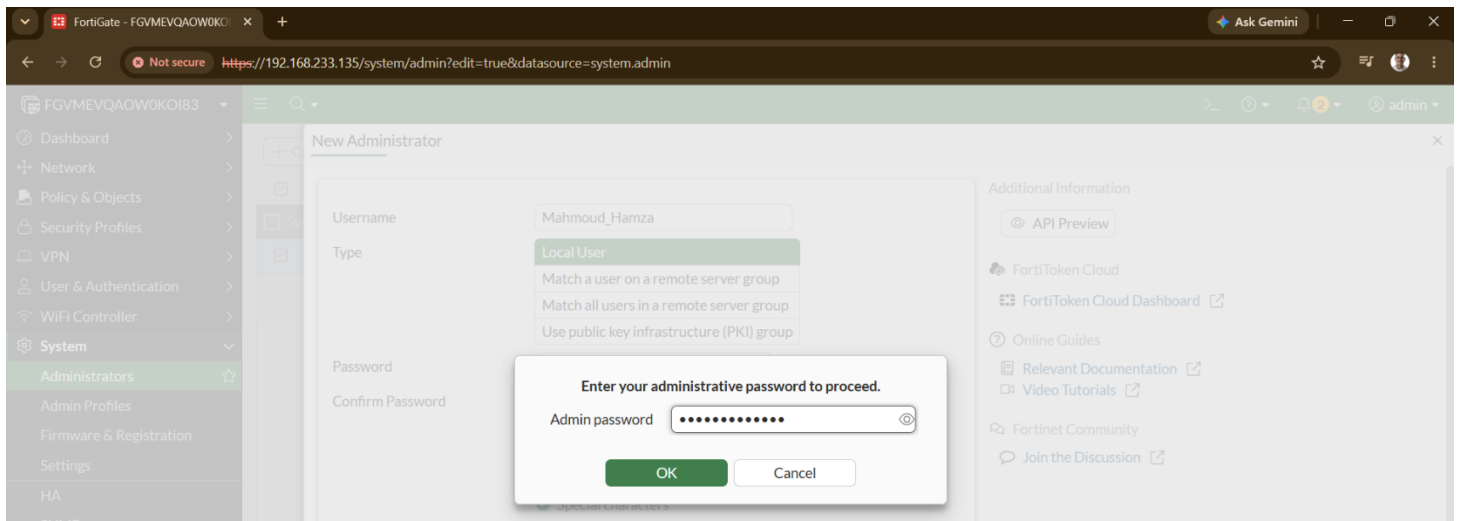


Figure 6.4: Administrative password re-authentication prompt during the creation of a super_admin account.

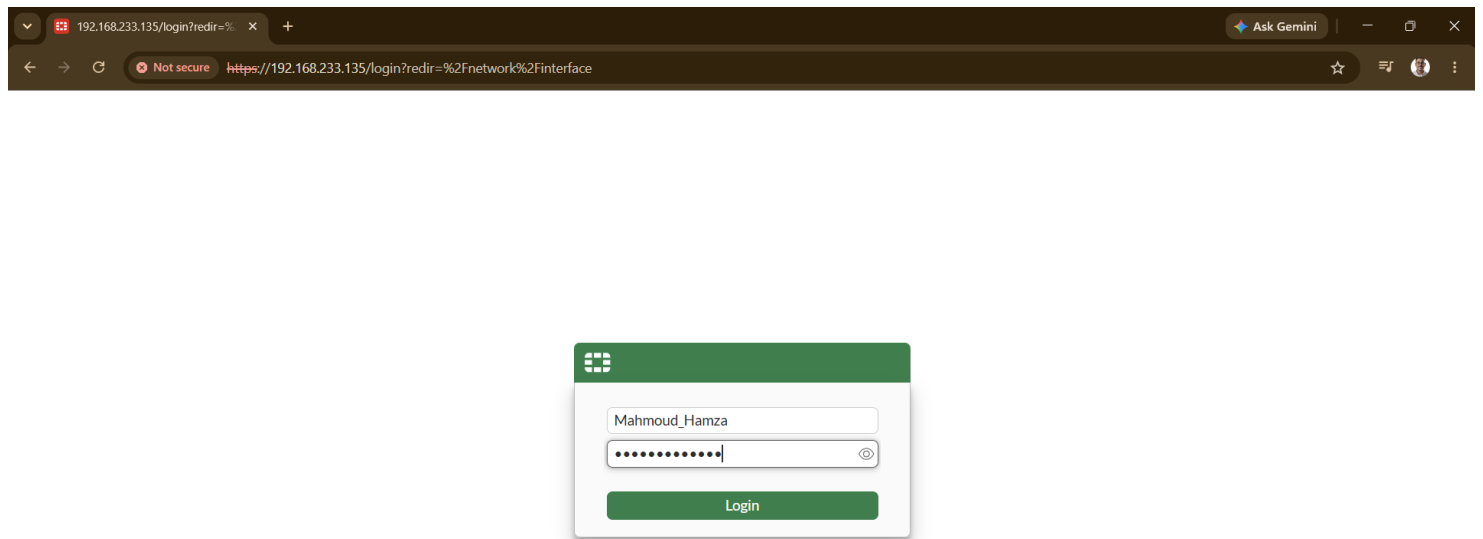


Figure 6.5: Establishing a new administrative session using the Mahmoud_Hamza credentials.

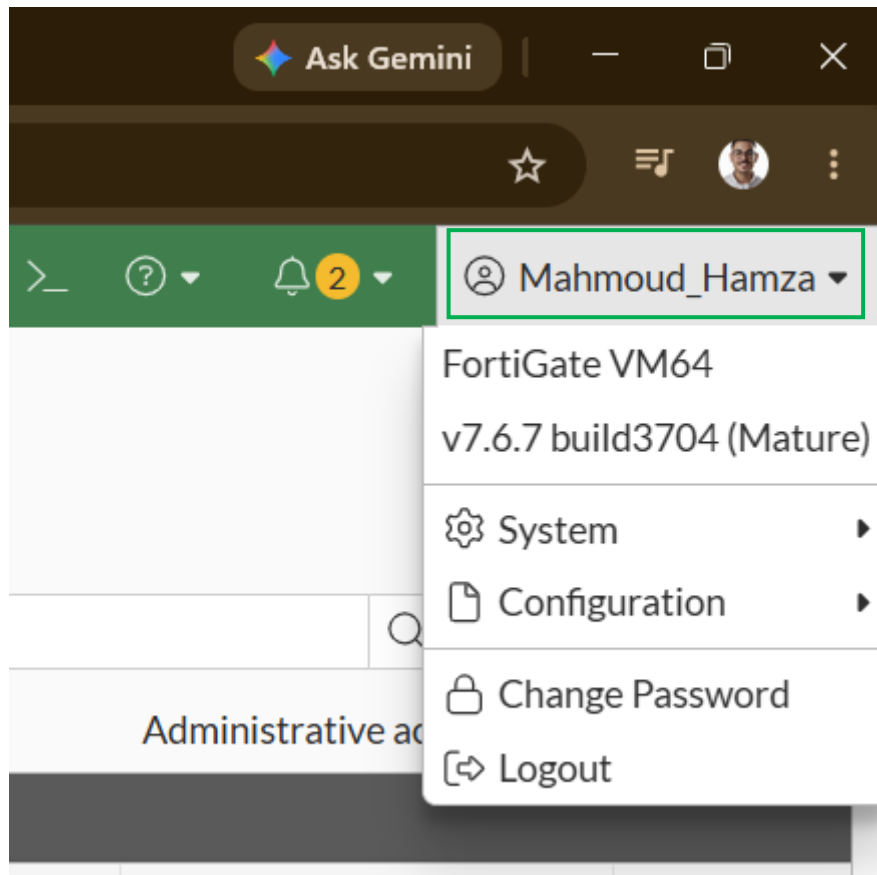


Figure 6.6: Verification of Active Administrative Session and Firmware Build Details (Mahmoud_Hamza - FortiGate VM64 v7.6.7).

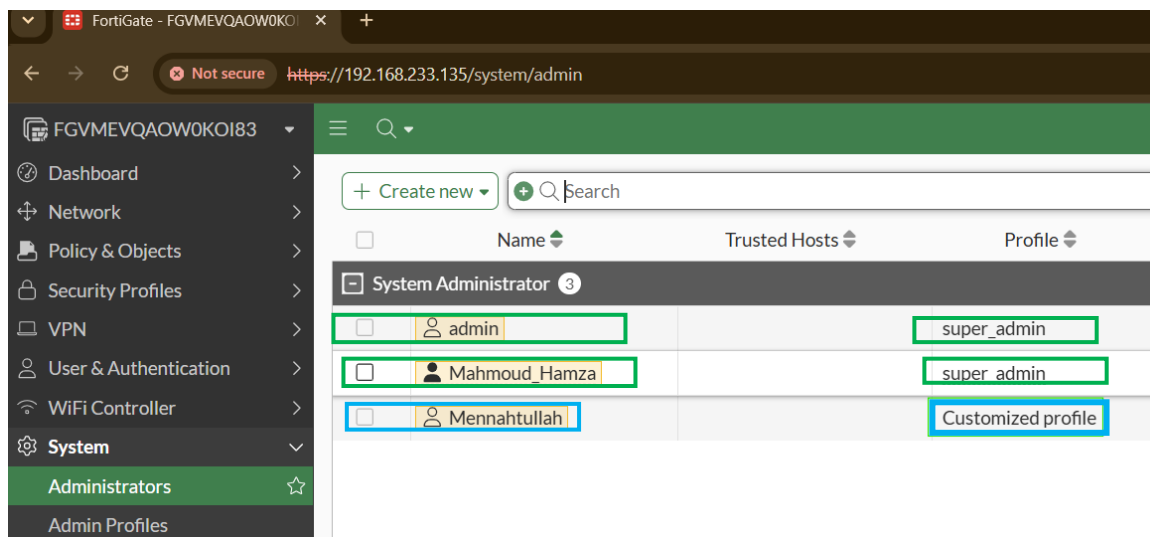


Figure 6.7: Complete System Administrators Table Displaying Super Admin Accounts (admin, Mahmoud_Hamza) and Restricted Operator Account (Mennahtullah) Assigned to Customized profile.

Step 6.3: Verification of Privilege Isolation & Role Boundaries

To confirm the operational enforcement of Role-Based Access Control (RBAC), authentication tests were conducted for both full-privilege and restricted administrative accounts:

- **Full Administration Session (Mahmoud_Hamza):** Logging in with the super_admin role granted full access to all system menus, configuration panels, and administrative user management tools.

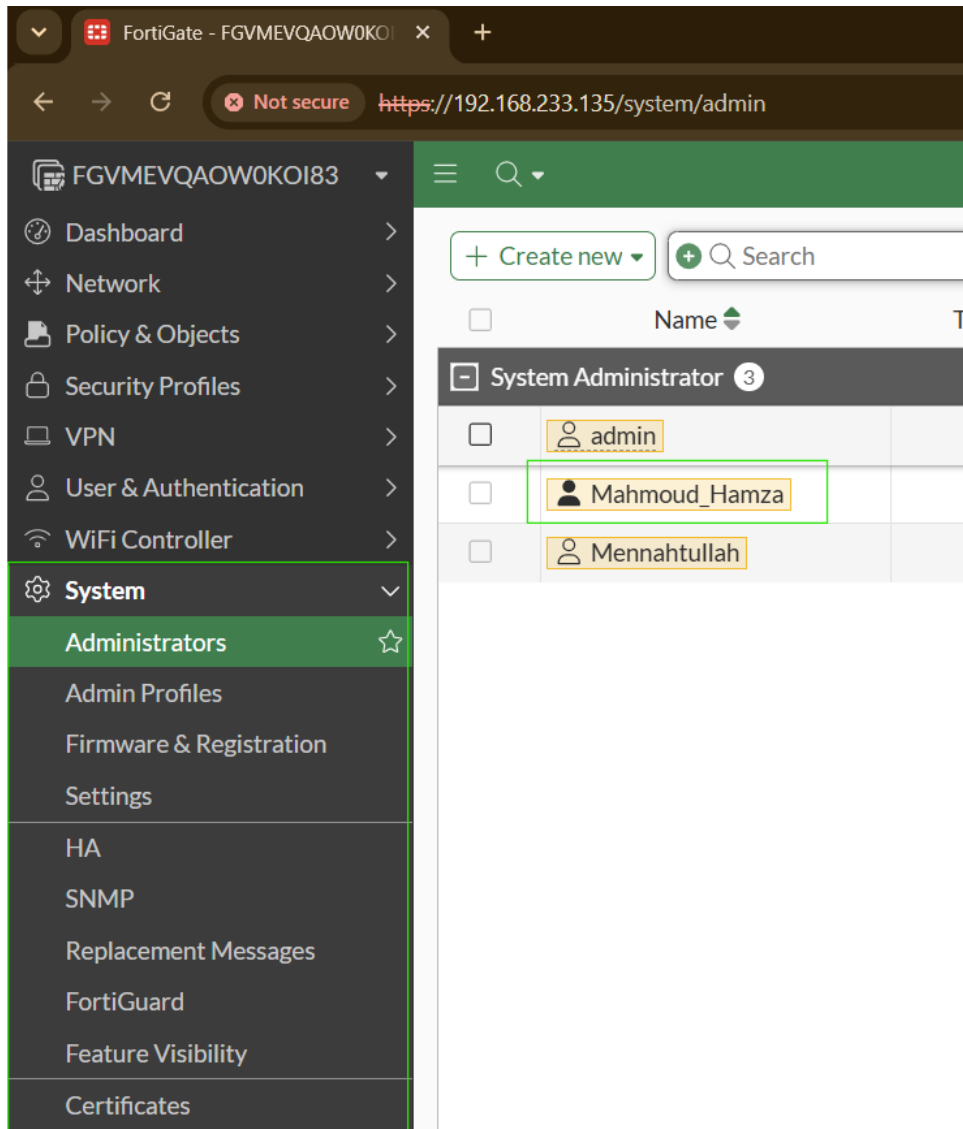


Figure 6.8: Active Super Admin Control-Plane Session under Mahmoud_Hamza, Demonstrating Unrestricted Access to System Menus and Configuration Options.

- **Restricted Operator Session (Mennahtullah):** Authenticating under the Customized profile successfully restricted operational actions. Menus were constrained to Read-Only status where configured, preventing unauthorized policy modifications or privilege escalation.



Figure 6.9: Administrative Web Portal Authentication Request Initiated for Restricted Account (Mennahtullah).

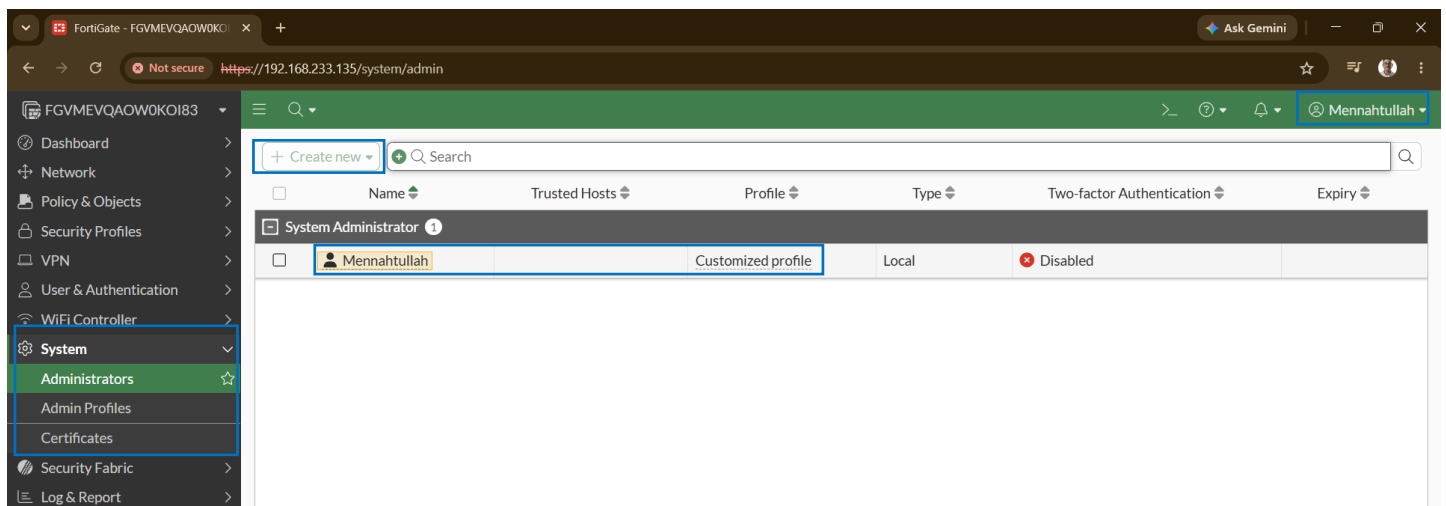


Figure 6.10: Active GUI Session under Restricted Operator (Mennahtullah), Demonstrating Scope Boundaries and Greyed-Out Administrative Functions (+ Create new Disabled).

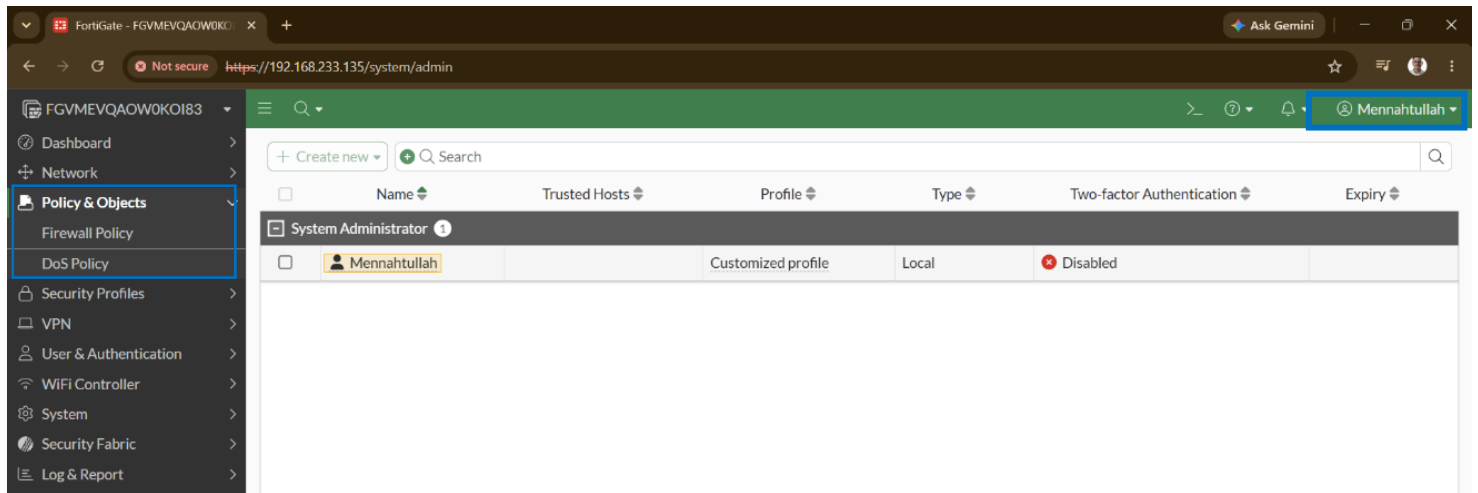


Figure 6.11: Role-Based Privilege Restrictions Applied under Mennahtullah (Customized profile), Restricting Policy & Objects Configuration Access.

7 :Firewall Address Objects Provisioning & Object Mapping

Step 7.1: Overview & Logical Abstraction Objectives

in this phase of the deployment, customized **Address Objects** were configured under **Policy & Objects > Addresses**. Abstracting network targets into logical Fully Qualified Domain Names (FQDN) and Geographical (GeoIP) definitions simplifies security management, enhances policy readability, and allows dynamic traffic enforcement without maintaining static IP lists.

Step 7.2: Executed Address Objects Configuration

To maintain an organized security posture and avoid hardcoding dynamic IP parameters into firewall rules, two distinct types of address objects were defined within FortiOS: **FQDN (Fully Qualified Domain Name)** and **Geography (GeoIP)**.

1. FQDN Object Provisioning (youtube_address):

Traditional IP-based objects are inefficient for modern web applications like YouTube that rely on multi-CDN infrastructure and rapidly shifting external IP ranges. By creating an FQDN object targeted directly at youtube.com with the interface set to any, FortiGate automatically monitors DNS queries, resolves the corresponding IP addresses via its internal dnsproxy engine, and maintains an updated dynamic translation table that can be globally referenced across multiple policies and interfaces.

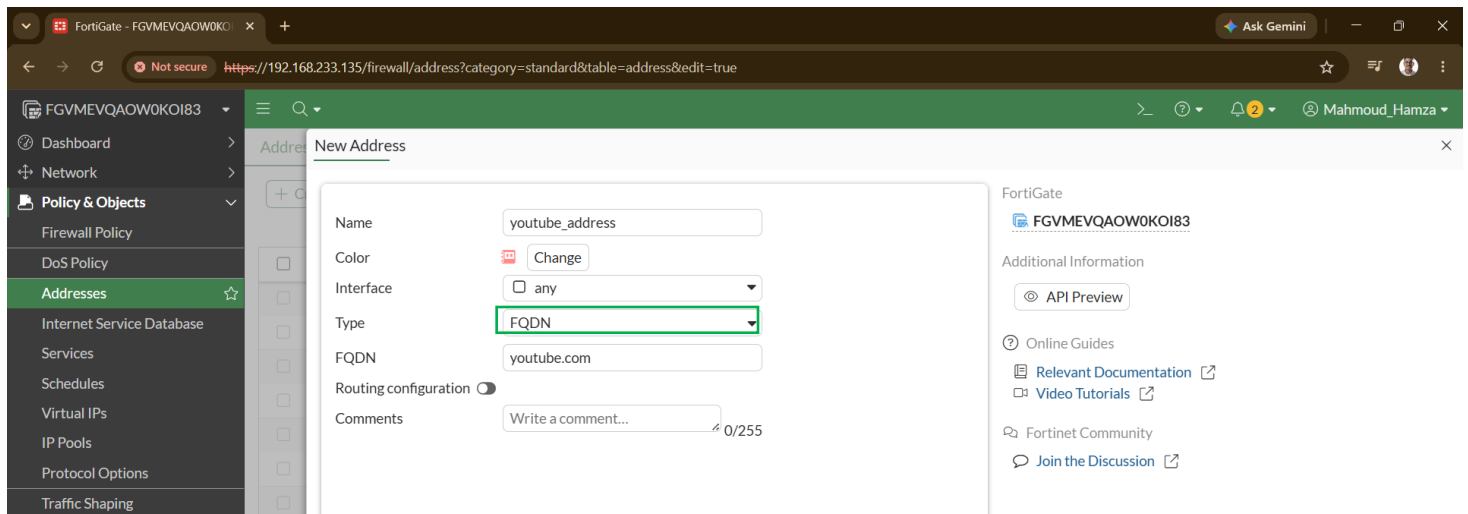


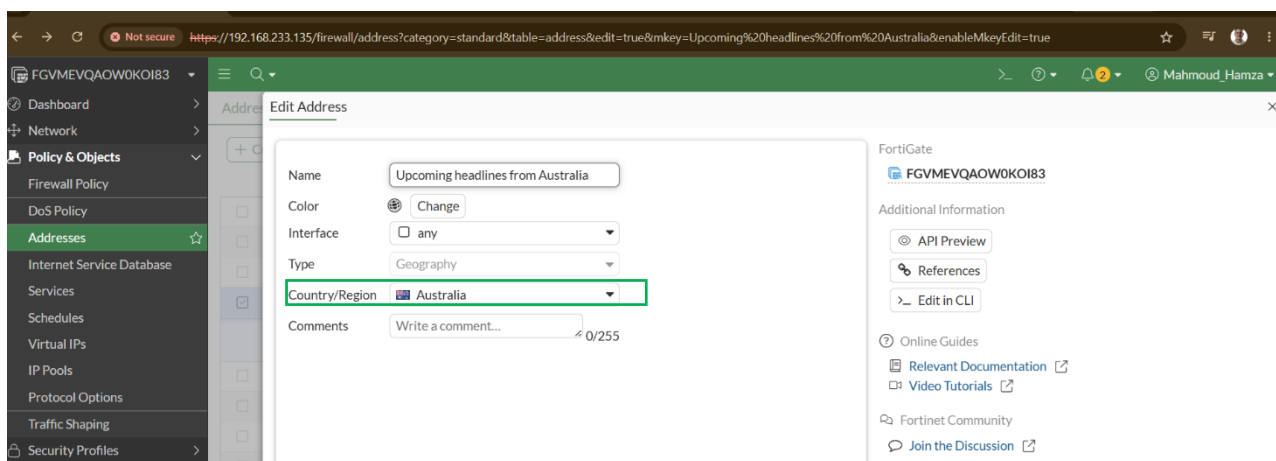
Figure 7.1: Creating the FQDN Address Object youtube_address mapped to youtube.com under any interface.

☒	youtube_address	FQDN	youtube.com	0
Edit Clone Delete More				

Figure 7.2: Verification of youtube_address FQDN entry successfully created in the Addresses Table.

2. Geographical Object Provisioning (Upcoming headlines from Australia):

GeoIP filtering allows policy controls based on geographical boundaries without needing manual maintenance of public IP ranges assigned to specific nations. By defining a Geography-type address object mapped to Australia, FortiGate interfaces with the dynamic FortiGuard GeoIP database. This allows administrators to construct location aware security policies (such as blocking or restricting regional traffic) seamlessly.



DOS Policy	☐	FABRIC_DEVICE	Subnet		0.0.0.0/0	0
Addresses	☐	FIREWALL_AUTH_PORTAL_ADDRESS	Subnet		0.0.0.0/0	0
Internet Service Database	☐	SSLVPN_TUNNEL_ADDR1	IP Range		10.212.134.200-10.212.134.210	0
Services	☐	Upcoming headlines from Australia	Geography	Australia		0
Schedules						

Figure 7.4: Final Addresses Table view displaying the registered Geography object for Australia.

Address Objects Summary

Address Name	Type	Value / Domain Target	Interface Binding	Security Purpose & Operational Scope
youtube_address	FQDN	youtube.com	any	Identifies destination web traffic targeting YouTube servers based on dynamic DNS resolution for domain-level access control across any interface.
Upcoming headlines from Australia	Geography	Australia	any	Leverages FortiGuard GeoIP classification to group external IP ranges assigned to Australia for regional traffic filtering.

8. Firewall Schedules, Services, and Security Policies Provisioning

Step 8.1: Time Based Schedule Provisioning

To enforce time based access control policies across network segments, customized Recurring Schedules were configured under Policy & Objects > Schedules. Defining distinct operational windows ensures network resources and internet capabilities are constrained strictly to authorized timeframes.

- Configuration & Implementation:**

Created a recurring schedule named Students to restrict connectivity to standard academic hours.

- Active Days:** Executed from Sunday through Thursday (excluding Friday and Saturday).
- Active Operational Window:** Configured from **08:00 AM to 02:00 PM (14:00)**.

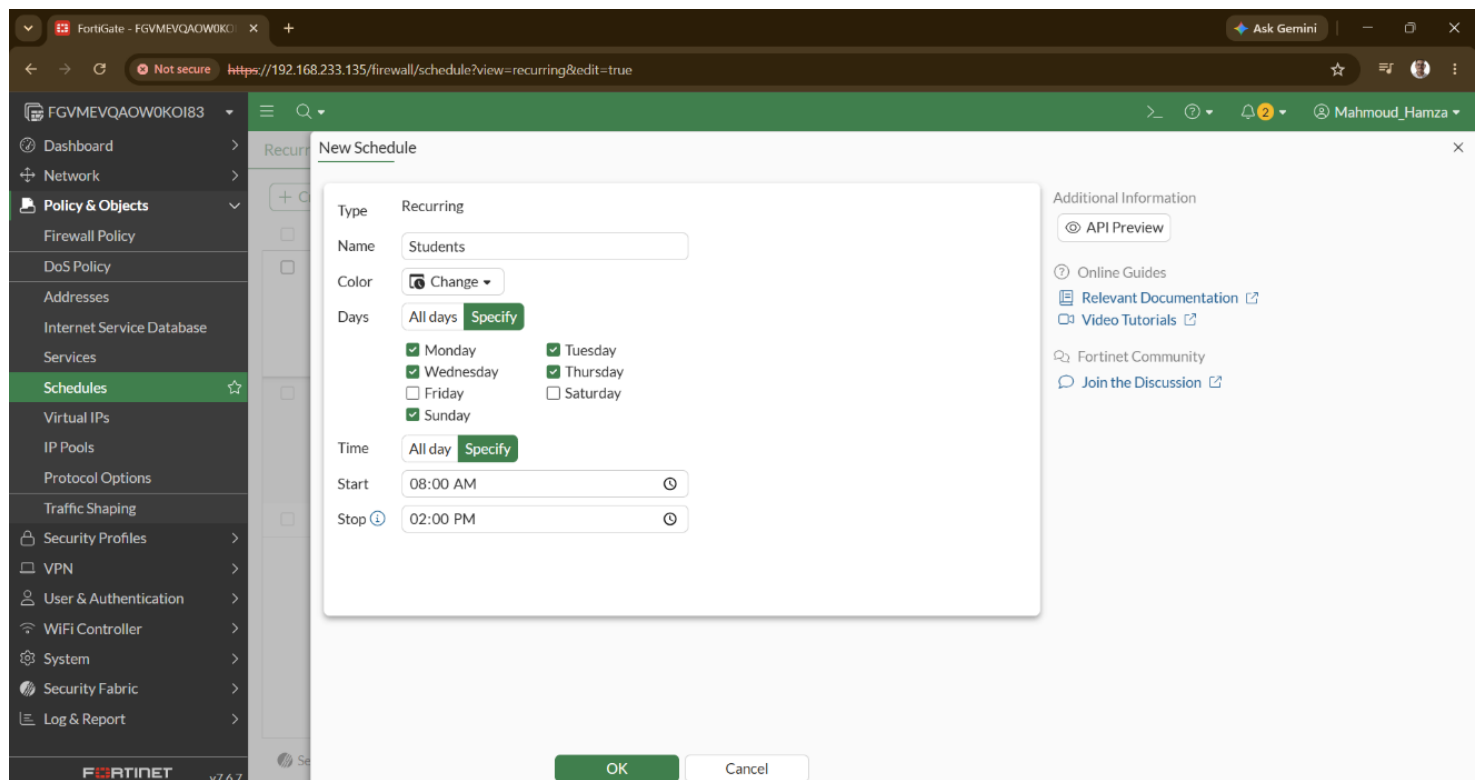


Figure 8.1: Configuring recurring schedule parameters for Students defining active days and operational hours.

	Name	Days	Start	End	Ref.
☐	Students	Sunday Monday Tuesday Wednesday Thursday	08:00:00	14:00:00	0
☐	always	Sunday Monday Tuesday Wednesday +3			2
☐	default-darrp-optimize	Sunday	01:00:00	01:30:00	1

Figure 8.2: Verification of registered Recurring Schedules displaying Students time definition and active policy references.

Step 8.2: Service Objects & Logical Group Consolidation

Under **Policy & Objects > Services**, default infrastructure services were inspected and bundled into structured **Service Groups**. Grouping application specific protocols streamlines policy creation and enforces granular access control over multi port services.

- 1. Service Database Verification:** Reviewed system default and custom services (totaling 85 defined services across Network, Remote Access, and File Access categories).
- 2. Service Groups Provisioning:** Inspected pre-configured and administrative service groups under **Service Group**:
 - **Email Access:** Consolidates mail transport protocols (DNS, IMAP, IMAPS, POP3).
 - **Web Access:** Bundles core web browsing and domain lookup protocols (DNS, HTTP, HTTPS).
 - **Exchange Server & Windows AD:** Groups directory and mail infrastructure protocols (DCE-RPC, DNS, HTTPS).

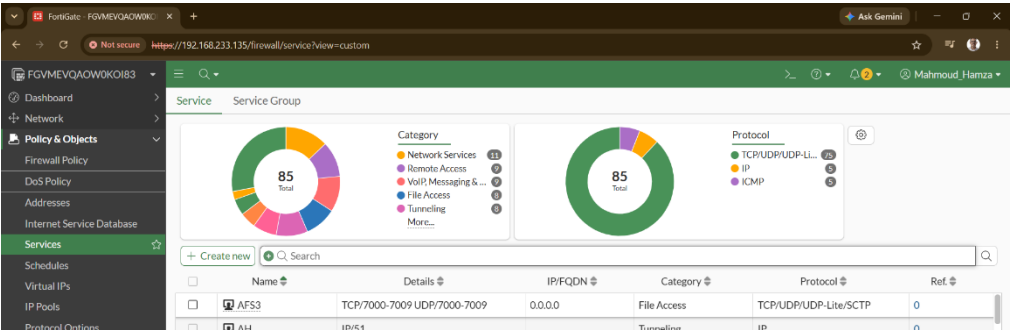


Figure 8.3: FortiOS Services Overview displaying active protocol classifications and network service categories.

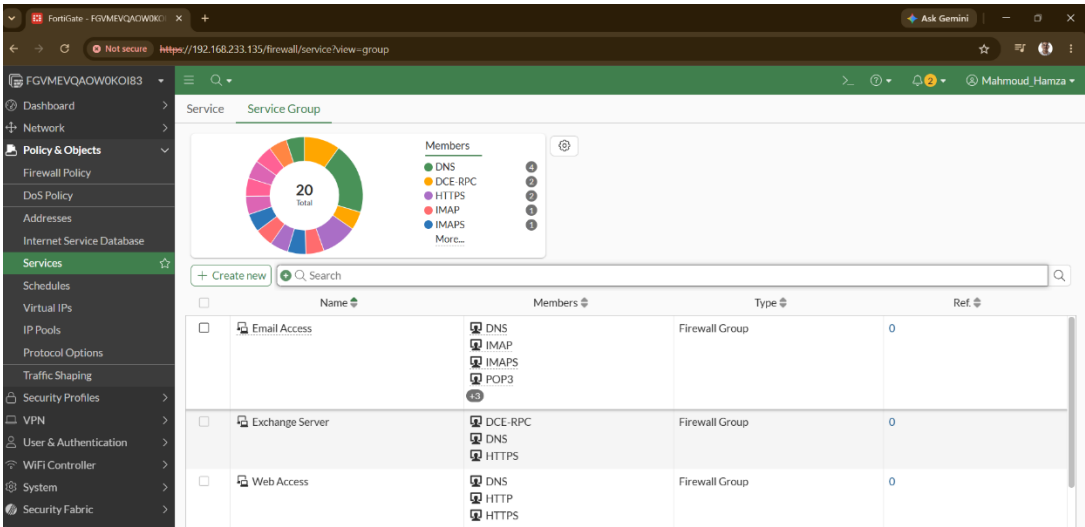


Figure 8.4: Custom Service Groups Table displaying protocol members assigned to Email Access, Web Access, and Directory Services.

9. Enterprise Firewall Policy Construction & Traffic Control

Step 9.1: Policy Rule Implementation (Students-Internet-Access)

To enforce security boundaries between internal assets and external networks, a granular firewall rule was created under **Policy & Objects > Firewall Policy**. The rule integrates the previously defined schedules, service groups, and network interfaces into a unified enforcement policy.

- **Execution Details:**

- **Policy Name:** Students-Internet-Access
- **Incoming Interface:** lan (port3) (Internal local network segment).
- **Outgoing Interface:** Wan (port2) (External internet gateway).
- **Source Scope:** port3 address (Local subnet host binding).
- **Destination Scope:** all (Outbound internet destinations).
- **Schedule Constraint:** Students (Restricted strictly to Sun–Thu, 08:00 AM – 02:00 PM).
- **Service Filtering:** Email Access and Web Access Service Groups.
- **Action & Inspection:** Set to **ACCEPT** with Network Address Translation (NAT) enabled and SSL Inspection set to no-inspection.

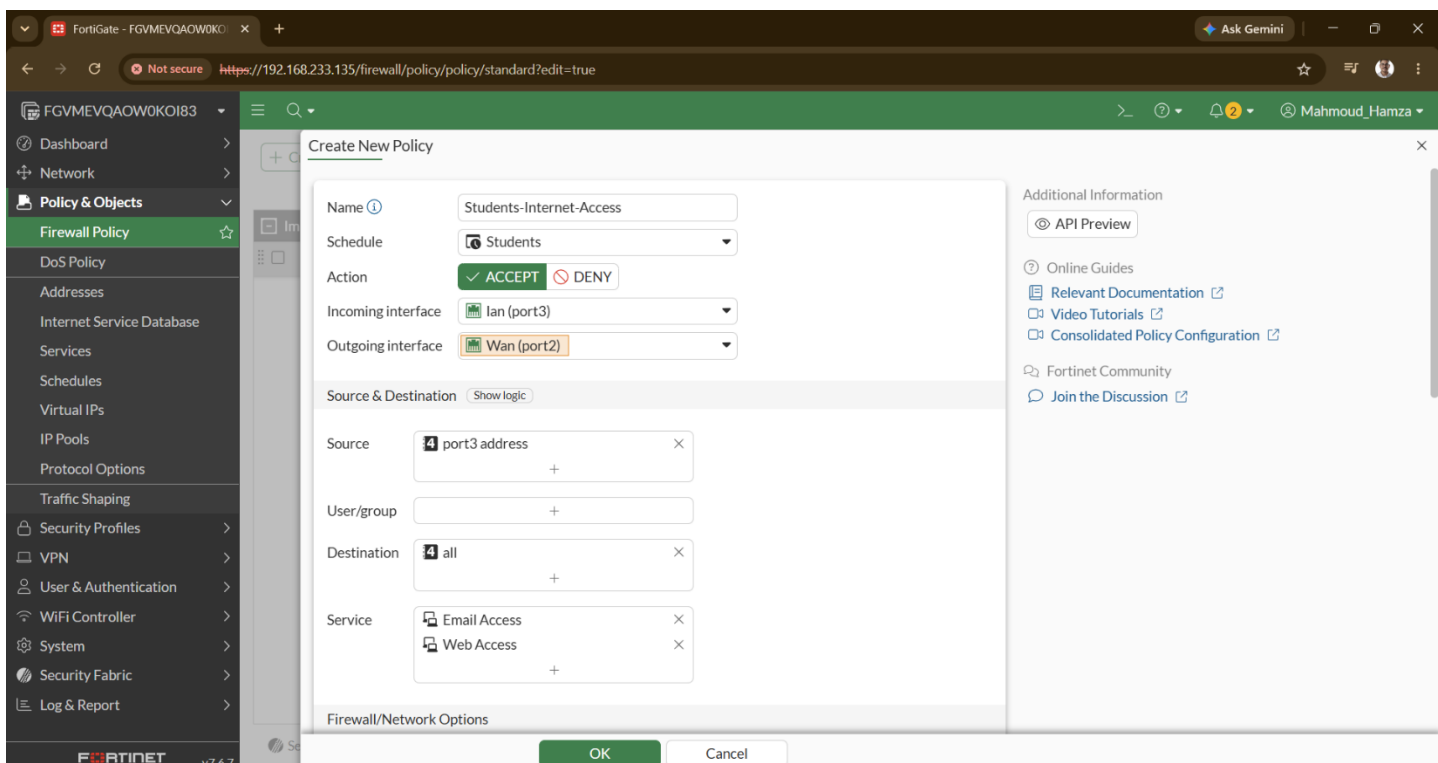


Figure 9.1: Configuring policy parameters for Students-Internet-Access binding interfaces, schedule constraints, and service groups.

The screenshot shows the FortiGate web interface for Firewall Policy configuration. The left sidebar lists navigation options: Dashboard, Network, Policy & Objects, Firewall Policy (selected), DoS Policy, and Addresses. The main area displays a table of firewall policies. The selected policy is 'Students-Internet-Access (1)'. The table columns are: Policy, Source, Destination, Schedule, Service, Action, IP Pool, NAT, Type, and Security Profile. The policy details are: Source is 'lan (port3) → Wan (port2)', Destination is 'port3 address → all', Schedule is 'Students', Service is 'Email Access, Web Access', Action is 'ACCEPT', NAT is 'NAT', Type is 'Standard', and Security Profile is 'no-insp'.

Policy	Source	Destination	Schedule	Service	Action	IP Pool	NAT	Type	Security Profile
Students-Internet-Access (1)	lan (port3) → Wan (port2)	port3 address → all	Students	Email Access, Web Access	ACCEPT		NAT	Standard	no-insp

Figure 9.2: Active Firewall Policy Table view verifying the enabled Students-Internet-Access rule and NAT translation.

Dynamic Time-Based Enforcement Mechanics:

The integration of the Students recurring schedule directly conditions the rule's operational state based on the FortiGate system clock:

- Authorized Window (Sun–Thu, 08:00–14:00):** FortiOS evaluates the policy as active, accepting and NATing allowed outbound web and email traffic.
- Non-Academic Hours & Weekends:** FortiOS automatically transitions the policy into an inactive state. Any egress traffic originating from student hosts matching this rule is immediately dropped by the firewall's lower-priority **Implicit Deny Policy**, effectively terminating internet connectivity during restricted hours.

Policy Name	Source Interface & Object	Destination Interface & Object	Schedule Constraint	Allowed Service Groups	Action & NAT
Students-Internet-Access	lan (port3) (port3 address)	Wan (port2) (all)	Students (Sun–Thu, 08:00–14:00)	Email Access, Web Access	ACCEPT (NAT Enabled)